

GNIOT Institute of Management Studies – PGDM (Batch 2024-26)**(TRIMESTER – III) END TERM EXAMINATION, MAY-2025****BASIC SQLFOR DATA ANALYTICS****[Time: 2:00 Hours]****[Maximum Marks: 60]****Following are the course outcomes of the subject: - Advance Excel for Managers**

CO	Course Outcome (CO)
CO1	Understand the fundamental elements of relational database management systems. Will be clear with the basic concepts of relational data model, entity-relationship model, relational database design, relational algebra and SQL.
CO2	Design ER-models to represent simple database application scenarios. Enhance the knowledge of the processes of Database Development and Administration using SQL.
CO3	Convert the ER-model to relational tables, populate relational databases and formulate SQL queries on data. Execute the commanding functions on various DB platforms.

SECTION-A**Attempt all parts of the following: (Short Answer Type Questions, word limit 100-150 words)**

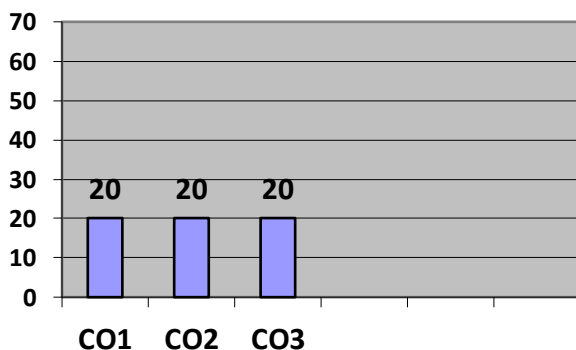
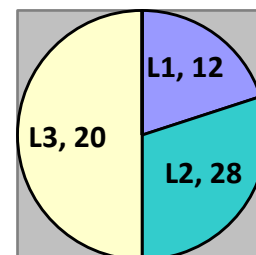
Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	4	CO – 1	BL-1
(B)	4	CO – 1	BL-2
(C)	4	CO – 1	BL-1
(D)	4	CO – 1	BL-2
(E)	4	CO – 1	BL-1

SECTION-B**Attempt Any two questions out of the following: (Long Answer Type Questions, word limit 200-250 words)**

Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	10	CO – 2	BL-2
2 (B)	10	CO – 2	BL-2
3 (A)	10	CO – 2	BL-3
3 (B)	10	CO – 2	BL-3

SECTION-C**Attempt all parts of the following: (Caselet / Numerical, word limit 200-250 words)**

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4(A)	10	CO – 3	BL-3
4(B)	10	CO – 3	BL-3

Course Outcome Wise Marks Distribution**L1 L2 L3****Bloom's Level wise Marks Distribution**

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GNIOT Institute of Management Studies – PGDM (Batch 2024-26)**(TRIMESTER – III) END TERM EXAMINATION, MAY - 2025****BASIC SQL FOR DATA ANALYTICS****[Time: 2:00 Hour]****[Maximum Marks: 60]****INSTRUCTIONS:**

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A**(Short Answer Type Questions, do not exceed 100 words)****Q.1. Attempt all parts of the following:****(5×4=20)**

- (a) What is the concept of database systems?
- (b) Explain relational database management systems and its applications.
- (c) What is the difference between traditional file management system and DBMS?
- (d) Explain ER model using an example.
- (e) How ER diagrams may be used in database design?

SECTION-B**(Long Answer Type Questions)****Attempt all questions of the following****(2×10=20)****Q.2(a)** Explain 3-tier architecture of DBMS, in detail.**OR****Q.2(b)** Explain the features of DBMS in detail.**Q.3(a)** Construct a comparison between sorting and filtering in SQL. How to use wildcard characters in filtering?**OR****Q3(b)** Apply the following queries on the given PRODUCT table.

- i) Sort the products from higher to lower Price.
- ii) Apply filter on ProductID in ascending order.
- iii) Sort the records by the "ProductID" and the "ProductName" column.
- iv) Sort the records by the "CategoryID" in ascending and the "Price" in descending.

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
1	Chais	1	1	10 boxes x 20 bags	18
2	Chang	1	1	24 - 12 oz bottles	19
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.35

SECTION-C

Q.4. Attempt all parts of the following:

(2×10=20)

4(a) Define Join in SQL. What are different types of joins, write SQL query also?

4(b) Apply SQL queries on given Customer table.

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden
6	Blauer See	Hanna Moos	Forsterstr. 57	Mannheim	68306	Germany

- i) Insert four records in the customer table.
- ii) Update the record having CustomerID = 3 with a new contact person *and* a new city.